

Loan APIs Technical Documentation

This document provides a comprehensive technical reference for the Loan APIs available in the LoanInNeed backend application. It details the endpoints, expected request formats, authentication requirements, and sample responses.

1. Apply for a Loan

Endpoint: `POST /api/loans/apply`

Description: Submits a new loan application for the authenticated user. It automatically captures any attribution parameters via middleware and triggers a background job to push the application to the external Loan Origination System (LOS).

Authentication: Required (`Bearer Token` in Authorization header)

Request Headers

Header	Type	Required	Description
<code>Authorization</code>	string	Yes	Format: <code>Bearer <jwt_token></code>
<code>Content-Type</code>	string	Yes	<code>application/json</code>

Request Body

Field	Type	Required	Description
<code>loanAmount</code>	number	Yes	The requested loan amount (e.g., <code>50000</code>).
<code>purposeOfLoan</code>	string	Yes	Reason or description for the loan.
<code>loanType</code>	string	Optional	Enum value for the type of loan. Default is <code>'OTHER'</code> . Examples: <code>'MEDICAL_EMERGENCY'</code> , <code>'EDUCATION'</code> , <code>'WEDDING'</code> , <code>'TRAVEL'</code> .

Sample Request (JSON):

```
{
  "loanAmount": 50000,
  "purposeOfLoan": "Medical Emergency for family",
  "loanType": "MEDICAL_EMERGENCY"
}
```

Response

Success Response (201 Created): Returned when the application is successfully created in the local database and the LOS integration background job is queued.

Sample Response Body:

```
{
  "message": "Loan application submitted successfully.",
  "applicationId": 105,
  "attribution": "Partner 12"
}
```

(Note: If no attribution partner exists, "attribution": "Organic" will be returned)

Error Responses:

- **401 Unauthorized:** Missing or invalid Bearer token.

```
{
  "message": "Not authorized, token failed"
}
```

- **400 Bad Request:** Missing required parameters or invalid data types.

```
{
  "message": "Validation Error: loanAmount is required"
}
```

Workflow Sequence Details

When `/api/loans/apply` is called, the backend follows this internal workflow:

1. **Attribution Check:** The system looks for active `partnerId` values locked to the user or passed within the current session to ensure proper affiliate/partner tracking (`attributionSource`).
2. **Database Persistence:** A new `LoanApplication` record is saved to the local database with status `PENDING`.
3. **LOS Queueing:** The system creates a `LosIntegrationJob` record. The `losWorker.js` cron job processes this in the background, transforming the data according to `losMapping.js` and securely POSTing it to the third-party LOS API.
4. **Activity Logging:** An `AttributionLog` interaction event is recorded referencing the partner (if applicable).

Future Endpoints (Planned API Stubs)

(Note: As the application expands, the following `GET` APIs are recommended to be added to `LoanRoutes.js` to support client-side tracking functionality.)

2. Get User Loans (Stub)

Endpoint: `GET /api/loans` **Description:** Fetch all loan applications made by the authenticated user. **Auth:** Required

3. Get Loan details by ID (Stub)

Endpoint: `GET /api/loans/:id` **Description:** Get specific tracking details and LOS status of a single loan. **Auth:** Required